

Prasham Soni

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Skills

Programming Languages: Python, C++, MATLAB, SQL, Arduino C/C++, ROS 2

Robotics Tools: PyTorch, OpenCV, CUDA, Mech-Mind Vision, Keyence Vision, Git, Linux, AWS, FFmpeg, RTSP/HLS, Issac Sim, Gazebo

Hardware: FANUC, KUKA, Keyence CV-X/XG, LiDAR, ESP32, Arduino, Raspberry Pi, UWB

Automation & Control: Modbus, EtherNet/IP, EtherCAT, OPC UA, Industrial Drives, Servo Systems, Ignition HMI, PLC Programming

Experience

Tesla, Inc

Jan 2026 – Present

Robotics Intern (Fremont, CA)

- Implemented a vision-based 3D perception and robot-guidance pipeline for autonomous pick-and-place, combining deep-learning object detection, 3D localization, pose estimation, and hand-eye calibration to correct robot offsets, achieve ± 0.2 mm placement accuracy, and sustain >98% first-pass yield.
- Executed commissioning and production-readiness validation for 30+ FANUC and KUKA robots, performing on-floor robot programming, motion verification, safety checks, and validation of protected-space zones, speed limits, interlocks, and tool behavior to enable safe sign-off and reliable line launch.
- Resolved robot programming and system-integration issues during deployment by debugging robot logic, cell interfaces, and integration faults, improving first-pass commissioning success and reducing bring-up rework.

Rivian Automotive, Inc.

May 2025 – Dec 2025

Robotics & Computer Vision Engineering Co-Op (Normal, IL)

- Developed a real-time RAFT/MemFlowNet-based optical-flow motion perception system (PyTorch, OpenCV, CUDA on AWS) for tugger/AGVs; achieved 48% faster inference throughput, enabled automated motion classification, downtime tracking, and event-type subclassification for root-cause analysis.
- Built a vision-guided robotic manipulation pipeline for FANUC arms using Python and Keyence multi-camera systems; implemented 3D object localization, camera robot frame transformations, and motion execution for precise robotic handling.

Keepsake Automation

Feb 2024 – Jun 2024

Computer Vision & Robotics Integration Intern (Ahmedabad, GJ)

- Developed real-time vision pipelines for object detection and pose-based alignment using Python and OpenCV, enabling autonomous robotic screwing and pick-and-place in a closed-loop automation system.
- Integrated perception outputs with KUKA robots through Beckhoff TwinCAT to support vision-guided motion and process execution.

ICUBE Technologies

Jan 2023 – May 2023

Robotics Research and Development Intern (Ahmedabad, GJ)

- Developed real-time vehicle tracking algorithms using Hector SLAM and ROS 2, generating bitmap-based scene representations to optimize storage, reduce data size, and extend retention for toll collection and vehicle profiling systems.

Projects

EinsteinVision: Autonomous Driving Visualization | Python, PyTorch, YOLO, RAFT, Blender ([Link](#))

Mar 2025 – Apr 2025

- Built a 3D autonomous-driving visualization dashboard using YOLOv11, pose estimation, and RAFT optical flow to detect and render lanes, vehicles, pedestrians, traffic signs, and motion in Blender, improving scene understanding and human-robot interaction.

SfM/NeRF Pipeline | Python, PyTorch, MATLAB, OpenCV, SciPy ([Link](#))

Feb 2025 – Mar 2025

- Designed and implemented a complete 3D reconstruction pipeline combining classical Structure-from-Motion (SfM) and deep learning-based Neural Radiance Fields (NeRF) using Python, OpenCV, MATLAB, and PyTorch, enabling accurate camera pose estimation and photo-realistic novel view synthesis from 2D images.

AutoCalib: Camera Calibration | Python, PyTorch, OpenCV, SciPy, NumPy ([Link](#))

Jan 2025 – Feb 2025

- Created a camera calibration tool based on “Zhengyou Zhang’s” method with automatic checkerboard detection, sub-pixel corner refinement, and non-linear optimization to estimate intrinsics/extrinsics and correct lens distortion.

Autonomous Warehouse Management System | C++, Python, Arduino, ESP32 ([Link](#))

Apr 2021 – Feb 2022

- Engineered an autonomous warehouse management system with 4 mobile robots, using ARUCO-based localization, OpenCV-based object detection, and C++ control logic on ESP32/Arduino for real-time navigation and path planning.

Education

Worcester Polytechnic Institute (GPA : 3.85/4.0)

Aug 2024 – Aug 2026

Master of Science, **Robotics Engineering** (Massachusetts, USA)

Nirma University (GPA : 3.23/4.0)

Aug 2019 – May 2023

Bachelor of Technology, **Instrumentation and Control Engineering** (Gujarat, IN)

Publications

Conference Paper : Implementation of Master-Slave Communication Using MQTT Protocol ([Link](#))

July 2023

Springer: Next Generation Systems and Networks, Proceedings of BITS EEE CON 2022